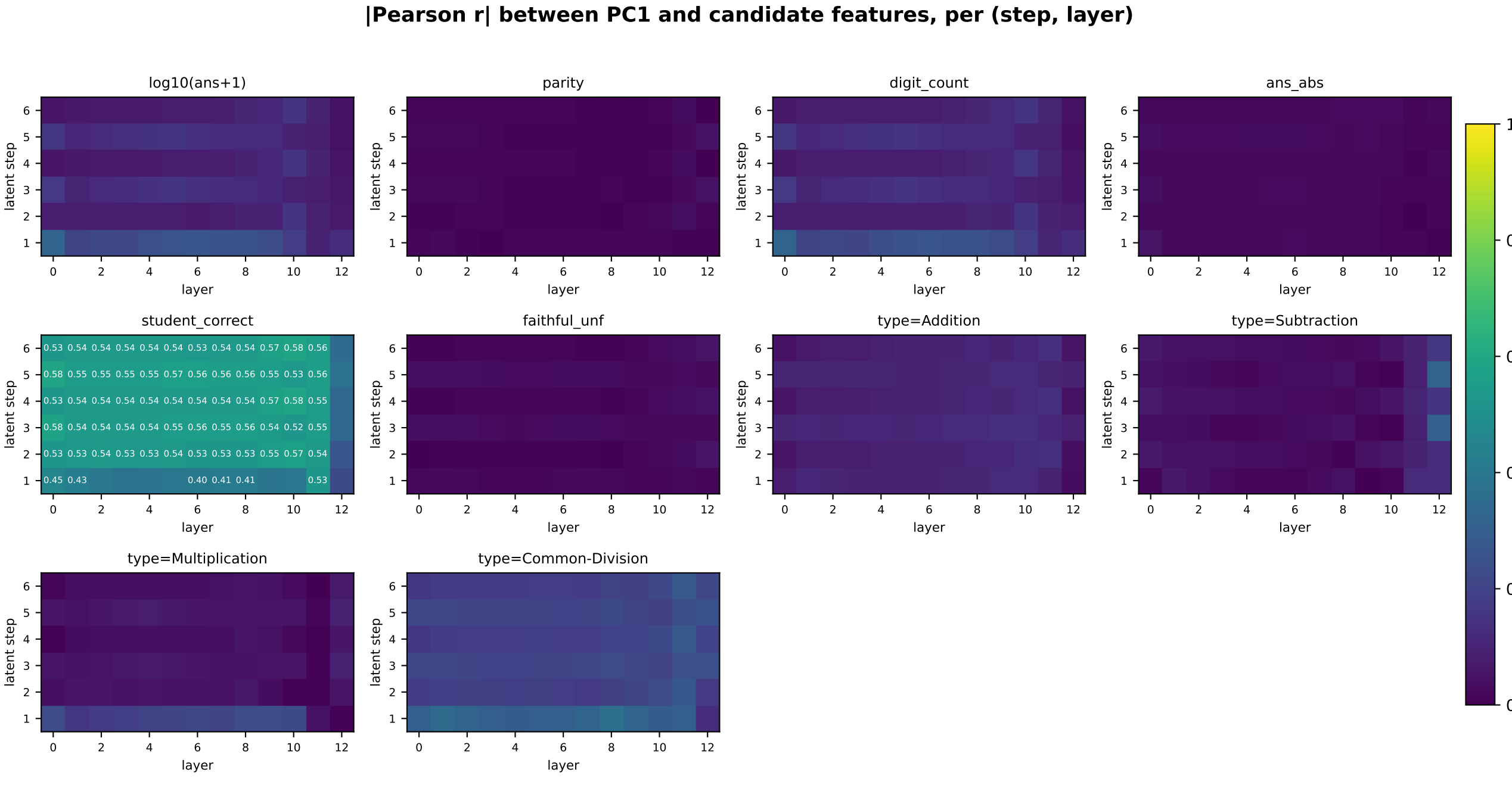
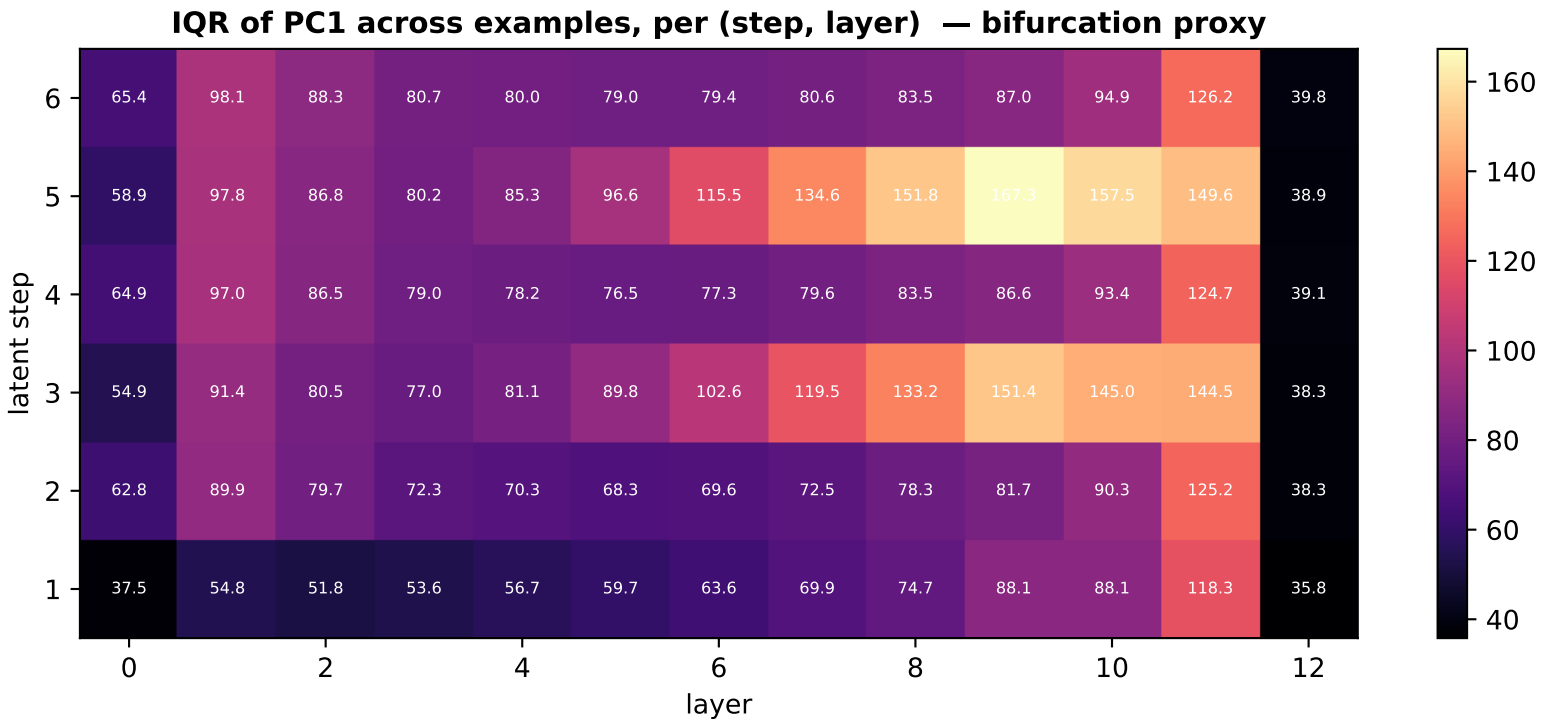
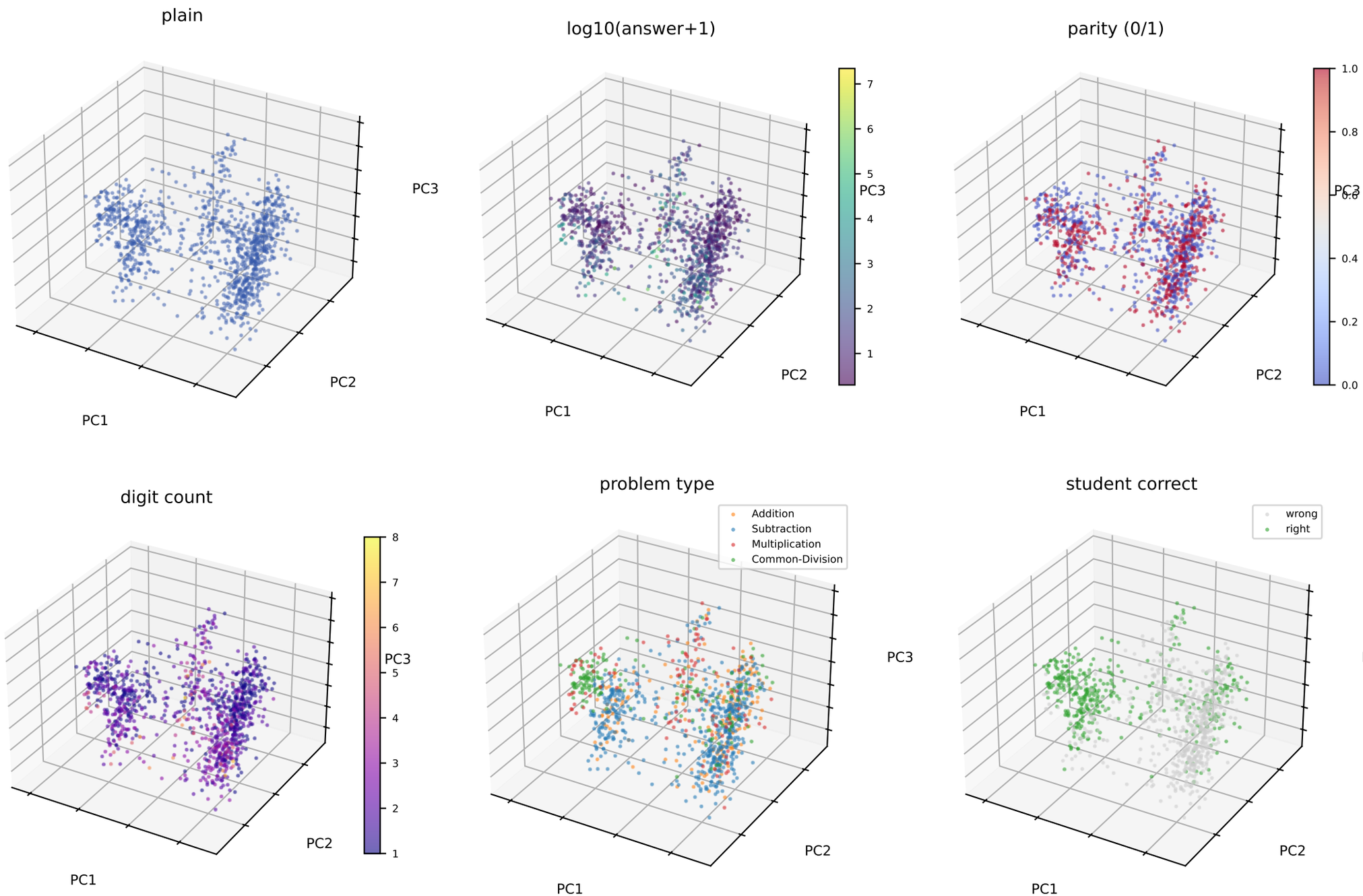
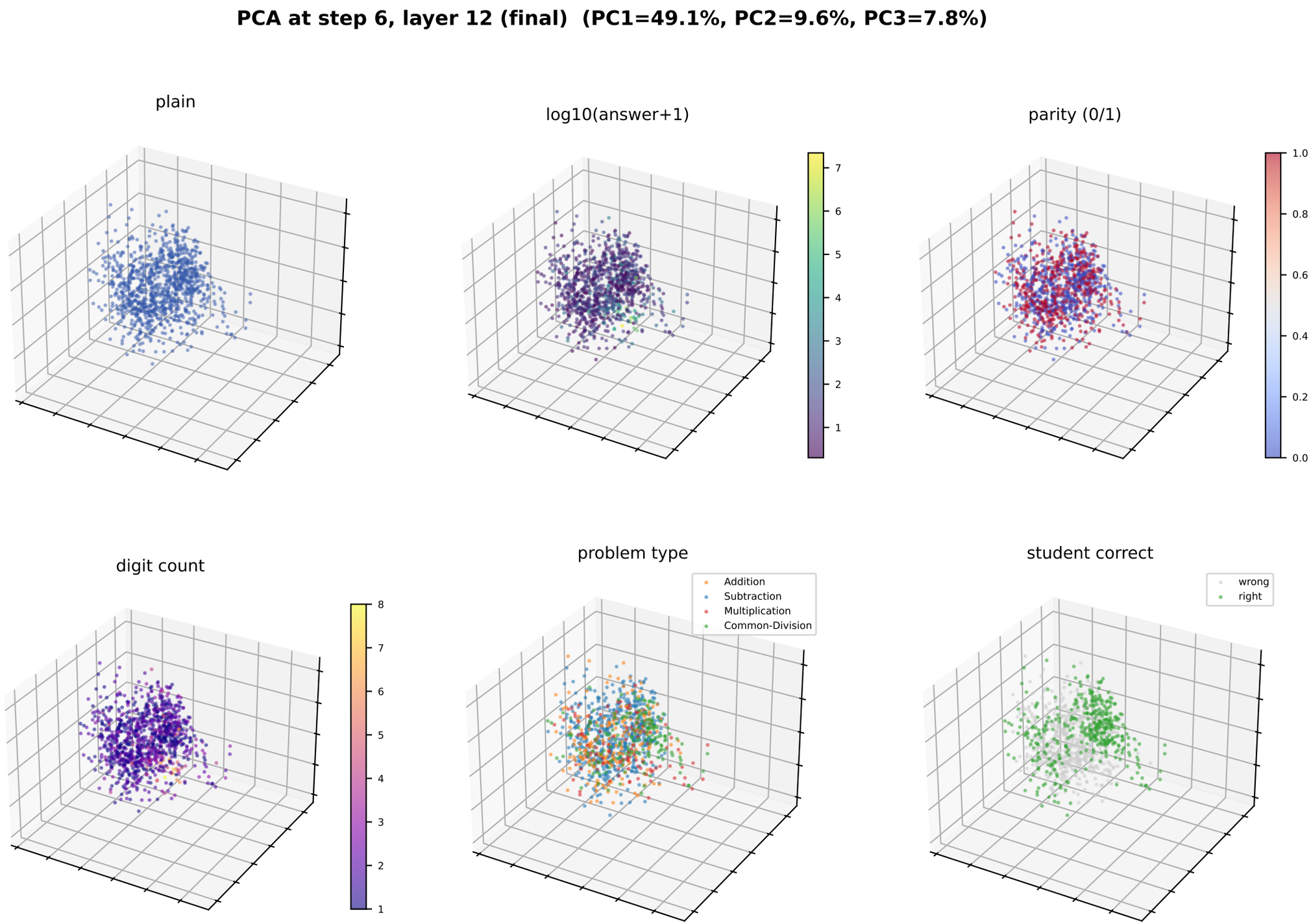

|Pearson r| between PC1 and candidate features, per (step, layer)
log10(ans+1)
parity
digit_count
ans_abs
student_correct
faithful_unf
type=Addition
type=Subtraction
type=Multiplication
type=Common-Division
latent step
layer
0.53 0.54 0.54 0.54 0.54 0.54 0.53 0.54 0.54 0.57 0.58 0.56
0.58 0.55 0.55 0.55 0.55 0.57 0.56 0.56 0.56 0.55 0.53 0.56
0.53 0.54 0.54 0.54 0.54 0.54 0.54 0.54 0.54 0.57 0.58 0.55
0.58 0.54 0.54 0.54 0.54 0.55 0.56 0.55 0.56 0.54 0.52 0.55
0.53 0.53 0.54 0.53 0.53 0.54 0.53 0.53 0.53 0.55 0.57 0.54
0.45 0.43 0.40 0.41 0.41 0.53


IQR of PC1 across examples, per (step, layer) — bifurcation proxy
latent step
layer
6 65.4 98.1 88.3 80.7 80.0 79.0 79.4 80.6 83.5 87.0 94.9 126.2 39.8
5 58.9 97.8 86.8 80.2 85.3 96.6 115.5 134.6 151.8 167.3 157.5 149.6 38.9
4 64.9 97.0 86.5 79.0 78.2 76.5 77.3 79.6 83.5 86.6 93.4 124.7 39.1
3 54.9 91.4 80.5 77.0 81.1 89.8 102.6 119.5 133.2 151.4 145.0 144.5 38.3
2 62.8 89.9 79.7 72.3 70.3 68.3 69.6 72.5 78.3 81.7 90.3 125.2 38.3
1 37.5 54.8 51.8 53.6 56.7 59.7 63.6 69.9 74.7 88.1 88.1 118.3 35.8
0 2 4 6 8 10 12
160 140 120 100 80 60 40

PCA of GPT-2 student acts at step 4, layer 10 (PC1=14.8%, PC2=6.3%, PC3=5.2%)
strongest PC1 driver: student_correct (|r|=0.585)




PCA at step 6, layer 12 (final) (PC1=49.1%, PC2=9.6%, PC3=7.8%)
plain
log10(answer+1)
parity (0/1)
digit count
problem type
student correct
Addition
Subtraction
Multiplication
Common-Division
wrong
right